MINI PROJECT 4

Chapter 2: Prerequisites & AWS Environment Setup

AWS CLI • Terraform • Ansible • SSH Key Pair

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Environment	Windows 11 with WSL2 Ubuntu (update if different)
Submission type	Local setup evidence; report actual AWS deployment status

Objective

To prepare the local WSL2 Ubuntu environment for infrastructure automation by installing and verifying AWS CLI, Terraform and Ansible, validating Terraform configuration, and generating an SSH key pair.

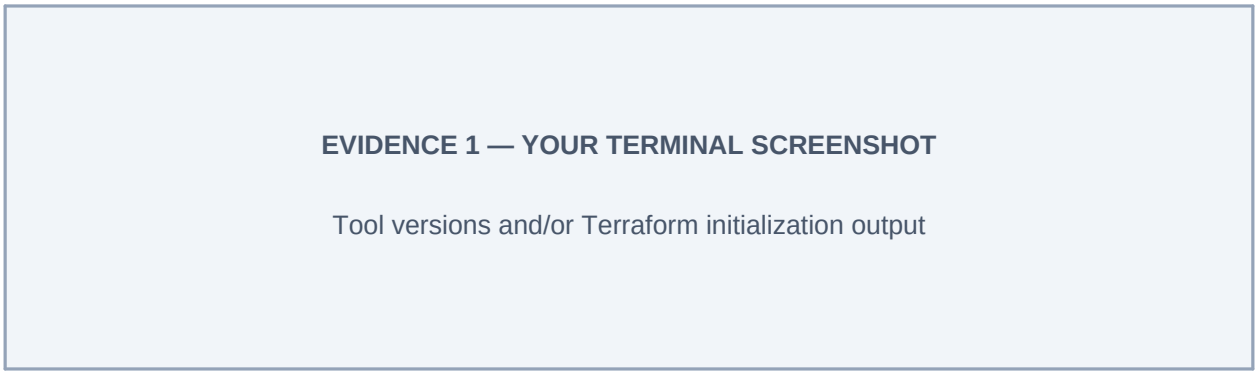
Scope note: Complete this report using your own terminal results. If AWS credentials are not configured or cloud resources were not created, mark those items as pending and do not claim a successful AWS deployment.

1. Tools and Configuration Verification

Run the relevant commands in your own WSL2 Ubuntu terminal. Capture screenshots showing the command and its actual output.

Commands you can use

```
aws --version
terraform -version
ansible --version
terraform init
terraform validate
```



Replace this box with a screenshot captured from your own terminal after running the relevant command.

2. Terraform Validation and Plan

Terraform initialization prepares the working directory and provider plugins. The validate command checks configuration structure. The plan command previews proposed changes; a plan can fail if provider credentials or other required inputs are missing.

EVIDENCE 2 — TERRAFORM VALIDATION

Capture the real output of: terraform validate

Replace this box with a screenshot captured from your own terminal after running the relevant command.

EVIDENCE 3 — TERRAFORM PLAN / ERROR, IF ANY

Capture the real output of: terraform plan. If it fails, describe the actual reason.

Replace this box with a screenshot captured from your own terminal after running the relevant command.

3. Terraform and Ansible Results

Record only results you personally obtained. Terraform init and validate do not prove that cloud resources were deployed. A plan that fails due to missing AWS credentials should be reported as incomplete.

EVIDENCE 4 — YOUR TERRAFORM / ANSIBLE OUTPUT

Use your actual command output. Do not paste another student's terminal screenshot.

Replace this box with a screenshot captured from your own terminal after running the relevant command.

Result summary

Terraform init: _____
Terraform validate: _____
Terraform plan: _____
Ansible version / check: _____

4. SSH Key Pair and AWS Authentication

An SSH key pair contains a private key and a public key. Keep the private key secret and never upload it to GitHub. On Linux, the private key should have restrictive permissions, commonly chmod 400.

Example commands (use your own project key name)

```
ssh-keygen -t ed25519 -f ~/.ssh/mahesh-aws-key
chmod 400 ~/.ssh/mahesh-aws-key
ls -l ~/.ssh/mahesh-aws-key*
aws sts get-caller-identity
```

EVIDENCE 5 — SSH KEY PERMISSIONS / ANSIBLE VERSION

Capture your actual terminal output. Never show private key contents.

Replace this box with a screenshot captured from your own terminal after running the relevant command.

EVIDENCE 6 — AWS IDENTITY CHECK

Capture the actual result of aws sts get-caller-identity; redact account details if needed.

Replace this box with a screenshot captured from your own terminal after running the relevant command.

5. Summary and Remaining Work

Update the status column after completing the tasks. This template does not assume your tools are installed or that AWS deployment was performed.

Item	Status
AWS CLI installed / version checked	Fill in actual status
Terraform installed / init / validate	Fill in actual status
Ansible installed / version checked	Fill in actual status
SSH key pair generated and permissions set	Fill in actual status
AWS credentials and STS identity	Fill in actual status
Public SSH key registered in AWS	Fill in actual status
Cloud infrastructure provisioned	Fill in actual status

Conclusion

Write a short conclusion based on your actual evidence. Clearly state which local setup tasks were completed and whether AWS authentication and cloud provisioning remain pending. Do not claim a successful deployment unless verified.

Security note: The private SSH key must remain confidential. Only the public key (.pub) should be registered with AWS.

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